

Web Project Final Evaluation

HerCircle - Requirement and Design Specification

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1. Introduction

1.1 Project Overview

HerCircle is a secure, MERN-stack-based web application designed as a social support platform exclusively for women. It enables users to engage in discussions around health, wellness, and personal topics in a safe and stigma-free environment.

1.2 Goals

- Provide a safe and supportive space for women
 - Enable anonymous sharing for sensitive topics
 - Foster community interaction through posts and discussions
 - Offer AI-powered guidance via an integrated chatbot
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2. Overall Description

2.1 User Characteristics

The target audience consists of women seeking support, guidance, or community discussions. This includes users looking for peer advice as well as those seeking private AI-based assistance. No technical expertise is required to use the platform.

2.2 Operating Environment

Component	Technologies / Stack
Frontend	React.js
Backend	Node.js, Express.js
Database	MongoDB
External API	Google Gemini API

3. Functional Requirements

3.1 User Authentication & Authorization

- Users must be able to sign up using email and password
 - Users must be able to log in and receive a JWT token
 - Protected routes must require valid authentication
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3.2 Community Feed & Posts

- Users can create text-based posts
 - Users can post anonymously
 - Posts can be categorized by topics such as PCOS, Cancer, Menstrual Health, Mental Health, Nutrition, Fitness, and Self-Defense
 - Users can view a chronological feed of posts
 - Users can filter posts based on selected topics
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3.3 Engagement Features

- Users can like posts
 - Users can comment on posts
 - Users can delete their own posts
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3.4 AI Chatbot (Gemini Integration)

- Users can access a private chatbot interface
- The system sends queries to the Gemini API and returns responses
- Chat history is saved and retrievable

4. Non-Functional Requirements

4.1 Security

- Passwords must be hashed using bcrypt
 - JWT-based authentication must be implemented
 - API endpoints must be secured
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4.2 Privacy

- Anonymous posting must be strictly enforced
 - User identity must not be exposed when anonymity is enabled
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4.3 Performance

- The community feed should load quickly
 - Efficient database queries should be used
 - Pagination or infinite scrolling should be implemented
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4.4 Responsiveness

- The application must be fully responsive
 - It should work seamlessly on mobile, tablet, and desktop devices
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5. System Architecture

HerCircle follows a Client-Server Architecture:

Client (Frontend):

Built using React.js, responsible for UI rendering, state management, and user interaction

Server (Backend):

Built using Node.js and Express.js, handles authentication, business logic, and API communication

Database:

MongoDB stores users, posts, comments, and chat histories

6. Database Schema Design

6.1 User Collection

Field Name	Data Type / Description
_id	ObjectId
username	String (Unique)
email	String (Unique)
password	String (Hashed)

6.2 Post Collection

Field Name	Data Type / Description
_id	ObjectId
author_id	ObjectId (Ref User)
content	String
isAnonymous	Boolean (default: false)
topic	String
likes	Array of ObjectIds
createdAt	Date

6.3 Comment Collection

Field Name	Data Type / Description
_id	ObjectId
author_id	ObjectId (Ref User)
content	String
isAnonymous	Boolean (default: false)
topic	String
likes	Array of ObjectIds
createdAt	Date

6.4 ChatHistory Collection

Field Name	Data Type / Description
_id	ObjectId
user_id	ObjectId (Ref User)
messages	Array of Objects {role, text, timestamp}

7. UI/UX Design Specifications

Layout Structure

Left Sidebar (Topics):

A navigation panel for browsing topics such as PCOS, Cancer, Menstrual Health, Reproductive Health, Mental Health, Nutrition, Fitness, and Self-Defense

Main Content Area (Community Feed):

Post Creation Box:

A text input area with an anonymous toggle and post button

Feed:

Each post displays:

- User name or "Anonymous"
- Timestamp
- Post content
- Like and comment buttons
- Topic tag

Chat Interface:

A dedicated chatbot window for private conversations with saved history

8. Timeline & Milestones

23rd March: Project name updated with repository link

26th March: Requirement and Design document uploaded

26th March: README completed with full project details

9th April (Final Deadline): Project to be completed with all deliverables uploaded

9. Future Scope

- User profile customization
 - Saved or bookmarked posts
 - Voice-based AI chatbot
 - Moderation and reporting system
 - Notification system
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10. Conclusion

HerCircle aims to create a safe, inclusive, and empowering digital space for women. By combining community-driven support with AI-powered assistance, the platform enhances accessibility, privacy, and overall well-being.